BSIT 2C TASK 1 - QO MEMBERS:

Paul Vincent A. Pajaganas

Lyxane C. Ganir

Trexie May D. Cruda

QUERY OPTIMIZATION – SLIDE 28-31

1. **Compare and contrast CHAR to VARCHAR datatypes?**

   CHAR and VARCHAR are both string data types, but they differ in how they store data. CHAR is a fixed-length datatype, meaning it always uses the full specified size, even if the actual data is shorter, which can waste storage space. On the other hand, VARCHAR is a variable-length datatype that only uses space based on the actual length of the data, making it more efficient. Based on the example shown, using VARCHAR is preferred over CHAR because it avoids unnecessary memory usage.

2. **When to use integer or numeric fields data type? And Why?**

   Integer or numeric data types should be used when the field contains only numeric information. This is because numeric types are more efficient and appropriate for storing numbers compared to character types like VARCHAR, which can waste space. As shown in the example, using INT for an employee ID is more efficient than using VARCHAR.

3. **What are the differences between UNION and UNION ALL?**

   UNION and UNION ALL both combine results from multiple queries. The difference is that UNION removes duplicate records, while UNION ALL keeps all records, including duplicates. Because UNION checks for duplicates, it is less efficient. Therefore, if there are no duplicate records, UNION ALL is preferred for better performance.

4. **What are the advantages in using the following**

   a. **DISTINCT –** One advantage of using DISTINCT is that it returns only unique or non-duplicate values from a specific column. It helps reduce unnecessary repeated data and makes queries more efficient when used only on needed columns rather than using DISTINCT on all columns. This improves query performance and promotes efficient data retrieval.

b. **LIMIT -** One advantage of using LIMIT is that it restricts the number of rows returned by a query, which helps improve efficiency and reduce unnecessary data retrieval. When only one result is needed, using LIMIT 1 can make the query faster and optimize performance by stopping the search once the required record is found. It also helps save system resources and makes data retrieval more efficient.

c. **!= operator -** Using alternatives to the != operator can improve query efficiency because it avoids full table scans. By separating conditions into specific comparisons like less than or greater than, the database can use indexes more effectively, resulting in faster performance and efficient data retrieval.

d. **<> operator -** Replacing the <> operator with separate comparison conditions helps optimize queries and reduce unnecessary scanning. This improves performance, allows better use of indexes, and makes retrieving data more efficient.

TASK 2:

Discuss how **EXPLAIN** command optimize the query.

The **EXPLAIN** command does not itself optimize queries it’s like a diagnostic tool. **EXPLAIN** command reveals why queries is slow so you will know what optimization do you need to apply